

MATHEMATICS

TARGET : JEE- Advanced

CAPS-10

LIMIT

SCQ (Single Correct Type) :

- If $[x]$ is the greatest integer less than or equal to x , then $\lim_{x \rightarrow 0} \left[\frac{x^2}{\sin x \tan x} \right] =$
 (A) -1 (B) 2 (C) 0 (D) 1
- If $\lim_{t \rightarrow x} \frac{e^t f(x) - e^x f(t)}{(t-x)(f(x))^2} = 2$ and $f(0) = \frac{1}{2}$, then find the value of $f'(0)$.
 (A) 4 (B) 2 (C) 0 (D) 1
- The value of $\lim_{x \rightarrow 0} \frac{(1+3x+2x^2)^{\frac{1}{x}} - (1+3x-2x^2)^{\frac{1}{x}}}{x}$ is _____.
 (A) $-e^3$ (B) $3e^2$ (C) $\frac{1}{3}e^{-2}$ (D) $4e^3$
- If $f(x) = \frac{\tan x}{x}$ then $\lim_{x \rightarrow 0} ([f(x)] + x^2)^{\frac{1}{\{f(x)\}}} =$ (where $[.]$ denotes the greatest integer function and $\{.\}$ denotes fractional part)
 (A) 3 (B) $\log 3$ (C) e^3 (D) Does not exist
- Let $f(x) = |\cos^{-1}(\sin x)| - |\sin^{-1}(\cos x)|$ and $\frac{3\pi}{2} \leq x \leq 2\pi$ $\lim_{x \rightarrow 2\pi} \frac{\sin x - \tan x}{[f(x)]^3} =$ _____.
 (A) $\frac{-1}{16}$ (B) $\frac{1}{32}$ (C) $\frac{-1}{8}$ (D) $\frac{1}{16}$
- The value of $\lim_{x \rightarrow 1} \left(\frac{p}{1-x^p} - \frac{q}{1-x^q} \right)$; $p, q \in \mathbb{N}$ equals
 (A) $\frac{p+q}{2}$ (B) $\frac{pq}{2}$ (C) $\frac{p-q}{2}$ (D) $\sqrt{\frac{p}{q}}$
- $\lim_{n \rightarrow \infty} \frac{1^2 n + 2^2 (n-1) + 3^2 (n-2) + \dots + n^2 \cdot 1}{1^3 + 2^3 + 3^3 + \dots + n^3}$ is equal to:
 (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{6}$
- The value of $\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{\cos^{-1}(2x\sqrt{1-x^2})}{x - \frac{1}{\sqrt{2}}}$ equals
 (A) $2\sqrt{2}$ (B) $-2\sqrt{2}$ (C) 0 (D) does not exist



9. The value of $\lim_{x \rightarrow \pi} \frac{1}{(x - \pi)} \left(\sqrt{\frac{4 \cos^2 x}{2 + \cos x}} - 2 \right)$ is
 (A) 0 (B) 2 (C) -2 (D) does not exist
10. For $n \in \mathbb{N}$, let x_n be defined as $\left(1 + \frac{1}{n}\right)^{n+x_n} = e$, then $\lim_{n \rightarrow \infty} x_n$ equals
 (A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{e}$ (D) 0

MCQ (One or more than one correct) :

11. If $\lim_{x \rightarrow 1} (2 - x + a[x - 1] + b[1 + x])$ exists, then a and b can take the values _____, (where $[]$ denotes the greatest integer function).
 (A) $a = \frac{1}{3}, b = 1$ (B) $a = 1, b = -1$ (C) $a = 9, b = -9$ (D) $a = 9, b = \frac{2}{3}$
12. If $A = \lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B = \lim_{x \rightarrow 0} \frac{[x]}{x}$, where $[]$ represents GIF, then _____.
 (A) $A = 1$ (B) A does not exist (C) $B = 0$ (D) $B = 1$
13. $\lim_{x \rightarrow \infty} \left[mx \sin \frac{1}{x} \right]$ is equal to (where $[.]$ denotes the greatest integer function and $m \in \mathbb{I}$)
 (A) m if $m < 0$ (B) $m - 1$ if $m > 0$ (C) $m - 1$ if $m < 0$ (D) m if $m > 0$
14. If $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function and $f(0) = 0$ and $f'(0) = 1$, then

$$\lim_{x \rightarrow 0} \frac{f(x) + f\left(\frac{x}{2}\right) + f\left(\frac{x}{3}\right) + \dots + f\left(\frac{x}{n}\right)}{x}$$
, where $n \in \mathbb{N}$, equals
 (A) 0 (B) $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$
 (C) ${}^nC_1 - \frac{{}^nC_2}{2} + \frac{{}^nC_3}{3} - \dots + (-1)^{n-1} \frac{{}^nC_n}{n}$ (D) does not exist

Comprehension Type Question:

Comprehension # 1

Let $f(x) = \frac{\sin^{-1}(1 - \{x\}) \times \cos^{-1}(1 - \{x\})}{\sqrt{2\{x\}} \times (1 - \{x\})}$ where $\{x\}$ denotes the fractional part of x .

15. $L = \lim_{x \rightarrow 0^+} f(x)$ is equal to
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$ (C) $\frac{\pi}{\sqrt{2}}$ (D) $\sqrt{2}\pi$
16. $R = \lim_{x \rightarrow 0^-} f(x)$ is equal to
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$ (C) $\frac{\pi}{\sqrt{2}}$ (D) $\sqrt{2}\pi$
17. Which of the following is not true ?
 (A) $\cos L < \cos R$ (B) $\tan(2L) > \tan R$ (C) $\sin L > \sin R$ (D) None of these

Comprehension # 2

Let $f(x) = \lim_{n \rightarrow \infty} \left(\cos \sqrt{\frac{x}{n}} \right)^n$, $g(x) = \lim_{n \rightarrow \infty} \left(1 - x + x \sqrt[n]{e} \right)^n$. Now, consider the function $y = h(x)$, where

$$h(x) = \tan^{-1} \left(g^{-1} f^{-1}(x) \right)$$

18. $\lim_{n \rightarrow 0^+} \frac{\ln(f(x))}{\ln(g(x))}$ is equal to _____.

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 0 (D) 1

19. Domain of the function $y = h(x)$ is _____.

(A) $(0, \infty)$ (B) $\mathbb{R}$ (C) $(0, 1)$ (D) $[0, 1]$

20. Range of the function $y = h(x)$ is _____.

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2}, 0\right)$ (C) $\mathbb{R}$ (D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Comprehension # 3

Column-I

(I) If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0$, then

(II) $\lim_{x \rightarrow 0} \left(1 + ax + bx^2 \right)^{\frac{2}{x}} = e^3$, then

(III) $\lim_{x \rightarrow 0} \left(\frac{ae^x - b}{x} \right) = 2$

(IV) $\lim_{x \rightarrow -\infty} \left\{ \sqrt{x^2 - x + 1} - ax - b \right\} = 0$

Column-II

(i) $a = 2$ (P) $b \in \mathbb{R}$

(ii) $a = \frac{3}{2}$ (Q) $b = 2$

(iii) $a = 1$ (R) $b = -1$

(iv) $a = -1$ (S) $b = \frac{1}{2}$

21. Which of the following is correct?

(A) (III) – (ii) – (R) (B) (III) – (iv) – (R) (C) (III) – (i) – (Q) (D) (III) – (ii) – (S)

22. Which of the following is correct?

(A) (II) – (ii) – (P) (B) (II) – (ii) – (R) (C) (II) – (iii) – (Q) (D) (II) – (iv) – (S)

23. Which of the following is correct?

(A) (I) – (ii) – (R) (B) (I) – (iii) – (R) (C) (I) – (iii) – (S) (D) (I) – (iv) – (P)

Comprehension # 4

Given $g(x) = \sin \frac{1}{x}$, $h(x) = x^2 + 2x + (\lambda + 1)$ and $u(x) = \frac{1}{x} + \cos \frac{1}{x^2}$. Let

$$f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n+1} g(x) + h(x)}{x^{2n} + 3x \cdot u(x)}, n \in \mathbb{I}$$

On the basis of above information, answer the following questions:

24. $\lim_{x \rightarrow \infty} f(x)$ is equal to :

(A) 0 (B) 2 (C) 5 (D) 1

25. If $\lim_{x \rightarrow 0} f(x) = 2$, then the value of λ is ;

(A) 10

(B) 5

(C) 2

(D) does not exist

Numerical based Questions :

26. If $\lim_{x \rightarrow 0} \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$ exists, then find the value of $a + b + c$

27. Let $K > 0$ and $\lambda = \lim_{n \rightarrow 0} \frac{K(1 - 4\sqrt{K^2 - x^2})}{x^2 \sqrt{K^2 - x^2}}$ is finite then the value of λK is

28. If $f(x) = \begin{cases} x-1, & x \geq 1 \\ 2x^2-2, & x < 1 \end{cases}$, $g(x) = \begin{cases} x+1, & x > 0 \\ -x^2+1, & x \leq 0 \end{cases}$ and $h(x) = |x|$,

then $\lim_{x \rightarrow 0} f(g(h(x)))$ is equal to :

29. Let $\lim_{x \rightarrow \infty} x \ln \left(e \left(1 + \frac{1}{x} \right)^{1-x} \right)$ equal $\frac{m}{n}$ where m and n are relatively prime positive integer.

Find $(m+n)$.

30. Let $f(x) = ax^3 + bx^2 + cx + d$ and $g(x) = x^2 + x - 2$.

If $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1$ and $\lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4$, then find the value of $\frac{c^2 + d^2}{a^2 + b^2}$

31. If $\lim_{x \rightarrow 3} \left(\frac{\sqrt{2x+3} - x}{\sqrt{x+1} - x + 1} \right)^{\frac{x-1-\sqrt{x^2-5}}{x^2-5x+6}}$ can be expressed in the form $\frac{a\sqrt{b}}{c}$ where $a, b, c \in \mathbb{N}$,

then find the least value of $(a^2 + b^2 + c^2)$.

32. Let $\lim_{x \rightarrow 1} \frac{x^6 + 1 + \frac{5}{3}(x^4 + x^2) - \frac{8}{3}(x^5 + x)}{x^5 + x + 6x^3 - 4x^4 - 4x^2} = \frac{p}{q}$ where $p, q \in \mathbb{N}$.

Find the smallest value of $(p+q)$.

Matrix Match Type :

33. Match the following

Column - I

(A) $\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x + \sin x \cos x}$

(B) $\lim_{x \rightarrow 0} \frac{\tan[-\pi^2]x^2 - [-\pi^2]x^2}{\sin^2 x}$

[.] denotes greatest integer function

(C) $\lim_{x \rightarrow 0} \left[\frac{x^2}{\sin x \tan x} \right]$

where [] denotes G.I. function

(D) $\lim_{x \rightarrow 0} \frac{2\sin^2 x}{2 - \sqrt{2(1 + \cos x)}}$

Column - II

(P) 8

(Q) 0

(R) 4

(S) -1

34. Column-I

Column-II

- (A) If $L = \lim_{x \rightarrow -1} \frac{\sqrt[3]{(7-x)} - 2}{(x+1)}$, then $12L =$ (P) -2
- (B) If $L = \lim_{x \rightarrow \pi/4} \frac{\tan^3 x - \tan x}{\cos\left(x + \frac{\pi}{4}\right)}$, then $\frac{-L}{4} =$ (Q) 2
- (C) If $L = \lim_{x \rightarrow 1} \frac{(2x-3)(\sqrt{x}-1)}{2x^2+x-3}$, then $20L =$ (R) 1
- (D) If $L = \lim_{x \rightarrow \infty} \frac{\log x^n - [x]}{[x]}$, where $n \in \mathbb{N}$, then $-2L =$ (S) -1

(Note : $[x]$ denotes greatest integer less than or equal to x)

35. Column-I

Column-II

- (A) If $\lim_{x \rightarrow \infty} (\sqrt{x^2 - x - 1} - ax - b) = 0$,
where $a > 0$, then there exists at least one
 a and b for which point $(a, 2b)$ lies on the line. (P) $y = -3$
- (B) If $\lim_{x \rightarrow 0} \frac{(1+a^3) + 8e^{1/x}}{1 + (1-b^3)e^{1/x}} = 2$, then there exists at least
one a and b for which point (a, b^3) lies on the line. (Q) $3x - 2y - 5 = 0$
- (C) If $\lim_{x \rightarrow \infty} (\sqrt{x^4 - x^2 + 1} - ax^2 - b) = 0$, then there exists
at least one a and b for which point $(a, -4b)$ lies on the line. (R) $15x - 2y - 11 = 0$
- (D) If $\lim_{x \rightarrow -a} \frac{x^7 + a^7}{x + a} = 7$, where $a < 0$, then there exists at least
one a for which point $(a, 2)$ lies on the line. (S) $y = 2$

ANSWER KEY OF CAPS-10

1. (C)	2. (C)	3. (D)	4. (C)	5. (D)
6. (C)	7. (A)	8. (D)	9. (A)	10. (B)
11. (BC)	12. (BC)	13. (AB)	14. (BC)	15. (A)
16. (B)	17. (B)	18. (B)	19. (C)	20. (D)
21. (C)	22. (A)	23. (B)	24. (D)	25. (B)
26. (0)	27. (2)	28. (0)	29. (5)	30. (16)
31. (29)	32. (13)	33. $A \rightarrow Q, B \rightarrow Q, C \rightarrow Q, D \rightarrow Q$		
34. $(A \rightarrow S; B \rightarrow R; C \rightarrow P; D \rightarrow Q)$	35. $(A \rightarrow Q; B \rightarrow P; C \rightarrow R, S; D \rightarrow S;$			